

Security+ CertMaster Study Course Outline



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1.0 Summarize Fundamental Security Concepts

- 1.1 Security Concepts
 - 1.1.1 Information Security
 - 1.1.2 Cybersecurity Framework
 - 1.1.3 Gap Analysis
 - 1.1.4 Access Control
- 1.2 Security Controls
 - 1.2.1 Security Control Categories
 - 1.2.2 Security Control Functional Types
 - 1.2.3 Information Security Roles and Responsibilities
 - 1.2.4 Information Security Competencies
 - 1.2.5 Information Security Business Units
- 1.3 Summary
- 1.4 Lesson Practice

2.0 Compare Threat Types

- 2.1 Threat Actors
 - 2.1.1 Vulnerability, Threat, and Risk
 - 2.1.2 Attributes of Threat Actors
 - 2.1.3 Motivations of Threat Actors
 - 2.1.4 Hackers and Hacktivists
 - 2.1.5 Nation-State Actors
 - 2.1.6 Organized Crime and Competitors
 - 2.1.7 Internal Threat Actors
- 2.2 Attack Surfaces
 - 2.2.1 Attack Surface and Threat Vectors
 - 2.2.2 Vulnerable Software Vectors
 - 2.2.3 Network Vectors
 - 2.2.4 Lure-Based Vectors
 - 2.2.5 Message-Based Vectors
 - 2.2.6 Supply Chain Attack Surface
- 2.3 Social Engineering
 - 2.3.1 Human Vectors
 - 2.3.2 Impersonation and Pretexting
 - 2.3.3 Phishing and Pharming
 - 2.3.4 Typosquatting
 - 2.3.5 Business Email Compromise
- 2.4 Summary
- 2.5 Lesson Practice

3.0 Explain Cryptographic Solutions

- 3.1 Cryptographic Algorithms
 - 3.1.1 Cryptographic Concepts
 - 3.1.2 Symmetric Encryption
 - 3.1.3 Key Length
 - 3.1.4 Asymmetric Encryption
 - 3.1.5 Hashing
 - 3.1.6 Digital Signatures
- 3.2 Public Key Infrastructure
 - 3.2.1 Certificate Authorities
 - 3.2.2 Digital Certificates
 - 3.2.3 Root of Trust
 - 3.2.4 Certificate Signing Requests
 - 3.2.5 Subject Name Attributes
 - 3.2.6 Certificate Revocation
 - 3.2.7 Key Management
 - 3.2.8 Cryptoprocessors and Secure Enclaves
 - 3.2.9 Key Escrow
- 3.3 Cryptographic Solutions
 - 3.3.1 Encryption Supporting Confidentiality
 - 3.3.2 Disk and File Encryption
 - 3.3.3 Database Encryption
 - 3.3.4 Transport Encryption and Key Exchange
 - 3.3.5 Perfect Forward Secrecy
 - 3.3.6 Salting and Key Stretching
 - 3.3.7 Blockchain
 - 3.3.8 Obfuscation
- 3.4 Summary
- 3.5 Lesson Practice

4.0 Implement Identity and Access Management

- 4.1 Authentication
 - 4.1.1 Authentication Design
 - 4.1.2 Password Concepts
 - 4.1.3 Password Managers
 - 4.1.4 Multifactor Authentication
 - 4.1.5 Biometric Authentication
 - 4.1.6 Hard Authentication Tokens
 - 4.1.7 Soft Authentication Tokens
 - 4.1.8 Passwordless Authentication
- 4.2 Authorization
 - 4.2.1 Discretionary and Mandatory Access Control
 - 4.2.2 Role- and Attribute-Based Access Control
 - 4.2.3 Rule-Based Access Control
 - 4.2.4 Least Privilege Permission Assignments
 - 4.2.5 User Account Provisioning
 - 4.2.6 Account Attributes and Access Policies
 - 4.2.7 Account Restrictions
 - 4.2.8 Privileged Access Management
- 4.3 Identity Management
 - 4.3.1 Local, Network, and Remote Authentication
 - 4.3.2 Directory Services
 - 4.3.3 Single Sign-on Authentication
 - 4.3.4 Single Sign-on Authorization
 - 4.3.5 Federation
 - 4.3.6 Security Assertion Markup Language
 - 4.3.7 Open Authorization
- 4.4 Summary
- 4.5 Lesson Practice

5.0 Secure Enterprise Network Architecture

- 5.1 Enterprise Network Architecture
 - 5.1.1 Architecture and Infrastructure Concepts
 - 5.1.2 Network Infrastructure
 - 5.1.3 Switching Infrastructure Considerations
 - 5.1.4 Routing Infrastructure Considerations
 - 5.1.5 Security Zones
 - 5.1.6 Attack Surface
 - 5.1.7 Port Security
 - 5.1.8 Physical Isolation
 - 5.1.9 Architecture Considerations
- 5.2 Network Security Appliances
 - 5.2.1 Device Placement
 - 5.2.2 Device Attributes
 - 5.2.3 Firewalls
 - 5.2.4 Layer 4 and Layer 7 Firewalls
 - 5.2.5 Proxy Servers
 - 5.2.6 Intrusion Detection Systems
 - 5.2.7 Next-Generation Firewalls and Unified Threat Management
 - 5.2.8 Load Balancers
 - 5.2.9 Web Application Firewalls
- 5.3 Secure Communications
 - 5.3.1 Remote Access Architecture
 - 5.3.2 Transport Layer Security Tunneling
 - 5.3.3 Internet Protocol Security Tunneling
 - 5.3.4 Internet Key Exchange
 - 5.3.5 Remote Desktop
 - 5.3.6 Secure Shell
 - 5.3.7 Out-of-Band Management and Jump Servers
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6.0 Secure Cloud Network Architecture

- 6.1 Cloud Infrastructure
 - 6.1.1 Cloud Deployment Models
 - 6.1.2 Cloud Service Models
 - 6.1.3 Responsibility Matrix
 - 6.1.4 Centralized and Decentralized Computing
 - 6.1.5 Resilient Architecture Concepts
 - 6.1.6 Application Virtualization and Container Virtualization
 - 6.1.7 Cloud Architecture
 - 6.1.8 Cloud Automation Technologies
 - 6.1.9 Software Defined Networking
 - 6.1.10 Cloud Architecture Features
 - 6.1.11 Cloud Security Considerations
- 6.2 Embedded Systems and Zero Trust Architecture
 - 6.2.1 Embedded Systems
 - 6.2.2 Industrial Control Systems
 - 6.2.3 Internet of Things
 - 6.2.4 Deperimeterization and Zero Trust
 - 6.2.5 Zero Trust Security Concepts
- 6.3 Summary
- 6.4 Lesson Practice

7.0 Explain Resiliency and Site Security Concepts

- 7.1 Asset Management
 - 7.1.1 Asset Tracking
 - 7.1.2 Asset Protection Concepts
 - 7.1.3 Data Backups
 - 7.1.4 Advanced Data Protection
 - 7.1.5 Secure Data Destruction
- 7.2 Redundancy Strategies
 - 7.2.1 Continuity of Operations
 - 7.2.2 Capacity Planning Risks
 - 7.2.3 High Availability
 - 7.2.4 Clustering
 - 7.2.5 Power Redundancy
 - 7.2.6 Diversity and Defense in Depth
 - 7.2.7 Deception Technologies
 - 7.2.8 Testing Resiliency
- 7.3 Physical Security
 - 7.3.1 Physical Security Controls
 - 7.3.2 Site Layout, Fencing, and Lighting
 - 7.3.3 Gateways and Locks
 - 7.3.4 Security Guards and Cameras
 - 7.3.5 Alarm Systems and Sensors
- 7.4 Summary
- 7.5 Lesson Practice

8.0 Explain Vulnerability Management

- 8.1 Device and OS Vulnerabilities
 - 8.1.1 Operating System Vulnerabilities
 - 8.1.2 Vulnerability Types
 - 8.1.3 Zero-Day Vulnerabilities
 - 8.1.4 Misconfiguration Vulnerabilities
 - 8.1.5 Cryptographic Vulnerabilities
 - 8.1.6 Sideloaded, Rooting, and Jailbreaking
- 8.2 Application and Cloud Vulnerabilities
 - 8.2.1 Application Vulnerabilities
 - 8.2.2 Evaluation Scope
 - 8.2.3 Web Application Attacks
 - 8.2.4 Cloud-based Application Attacks
 - 8.2.5 Supply Chain
- 8.3 Vulnerability Identification Methods
 - 8.3.1 Vulnerability Scanning
 - 8.3.2 Threat Feeds
 - 8.3.3 Deep and Dark Web
 - 8.3.4 Other Vulnerability Assessment Methods
- 8.4 Vulnerability Analysis and Remediation
 - 8.4.1 Common Vulnerabilities and Exposures
 - 8.4.2 False Positives, False Negatives, and Log Review
 - 8.4.3 Vulnerability Analysis
 - 8.4.4 Vulnerability Response and Remediation
- 8.5 Summary
- 8.6 Lesson Practice

9.0 Evaluate Network Security Capabilities

- 9.1 Network Security Baselines
 - 9.1.1 Benchmarks and Secure Configuration Guides
 - 9.1.2 Wireless Network Installation Considerations
 - 9.1.3 Wireless Encryption
 - 9.1.4 Wi-Fi Authentication Methods
 - 9.1.5 Network Access Control
- 9.2 Network Security Capability Enhancement
 - 9.2.1 Access Control Lists
 - 9.2.2 Intrusion Detection and Prevention Systems
 - 9.2.3 IDS and IPS Detection Methods
 - 9.2.4 Web Filtering
- 9.3 Summary
- 9.4 Lesson Practice

10.0 Assess Endpoint Security Capabilities

- 10.1 Implement Endpoint Security
 - 10.1.1 Endpoint Hardening
 - 10.1.2 Endpoint Protection
 - 10.1.3 Advanced Endpoint Protection
 - 10.1.4 Endpoint Configuration
 - 10.1.5 Hardening Techniques
 - 10.1.6 Hardening Specialized Devices
- 10.2 Mobile Device Hardening
 - 10.2.1 Mobile Hardening Techniques
 - 10.2.2 Full Device Encryption and External Media
 - 10.2.3 Location Services
 - 10.2.4 Cellular and GPS Connection Methods
 - 10.2.5 Wi-Fi and Tethering Connection Methods
 - 10.2.6 Bluetooth Connection Methods
 - 10.2.7 Near-Field Communications and Mobile Payment Services
- 10.3 Summary
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11.0 Enhance Application Security Capabilities

- 11.1 Application Protocol Security Baselines
 - 11.1.1 Secure Protocols
 - 11.1.2 Transport Layer Security
 - 11.1.3 Secure Directory Services
 - 11.1.4 Simple Network Management Protocol Security
 - 11.1.5 File Transfer Services
 - 11.1.6 Email Services
 - 11.1.7 Email Security
 - 11.1.8 Email Data Loss Prevention
 - 11.1.9 DNS Filtering
- 11.2 Cloud and Web Application Security Concepts
 - 11.2.1 Secure Coding Techniques
 - 11.2.2 Application Protections
 - 11.2.3 Software Sandboxing
- 11.3 Summary
- 11.4 Lesson Practice

12.0 Explain Incident Response and Monitoring Concepts

- 12.1 Incident Response
 - 12.1.1 Incident Response Processes
 - 12.1.2 Preparation
 - 12.1.3 Detection
 - 12.1.4 Analysis
 - 12.1.5 Containment
 - 12.1.6 Eradication and Recovery
 - 12.1.7 Lessons Learned
 - 12.1.8 Testing and Training
 - 12.1.9 Threat Hunting
- 12.2 Digital Forensics
 - 12.2.1 Due Process and Legal Hold
 - 12.2.2 Acquisition
 - 12.2.3 System Memory Acquisition
 - 12.2.4 Disk Image Acquisition
 - 12.2.5 Preservation
 - 12.2.6 Reporting
- 12.3 Data Sources
 - 12.3.1 Data Sources, Dashboards, and Reports
 - 12.3.2 Log Data
 - 12.3.3 Host Operating System Logs
 - 12.3.4 Application and Endpoint Logs
 - 12.3.5 Network Data Sources
 - 12.3.6 Packet Captures
 - 12.3.7 Metadata
- 12.4 Alerting and Monitoring Tools
 - 12.4.1 Security Information and Event Management
 - 12.4.2 Alerting and Monitoring Activities
 - 12.4.3 Alert Tuning
 - 12.4.4 Monitoring Infrastructure
 - 12.4.5 Monitoring Systems and Applications
 - 12.4.6 Benchmarks
- 12.5 Summary
- 12.6 Lesson Practice

13.0 Analyze Indicators of Malicious Activity

- 13.1 Malware Attack Indicators
 - 13.1.1 Malware Classification
 - 13.1.2 Computer Viruses
 - 13.1.3 Computer Worms and Fileless Malware
 - 13.1.4 Spyware and Keyloggers
 - 13.1.5 Backdoors and Remote Access Trojans
 - 13.1.6 Rootkits
 - 13.1.7 Ransomware, Crypto-Malware, and Logic Bombs
 - 13.1.8 TTPs and IoCs
 - 13.1.9 Malicious Activity Indicators
- 13.2 Physical and Network Attack Indicators
 - 13.2.1 Physical Attacks
 - 13.2.2 Network Attacks
 - 13.2.3 Distributed Denial of Service Attacks
 - 13.2.4 On-Path Attacks
 - 13.2.5 Domain Name System Attacks
 - 13.2.6 Wireless Attacks
 - 13.2.7 Password Attacks
 - 13.2.8 Credential Replay Attacks
 - 13.2.9 Cryptographic Attacks
 - 13.2.10 Malicious Code Indicators
- 13.3 Application Attack Indicators
 - 13.3.1 Application Attacks
 - 13.3.2 Replay Attacks
 - 13.3.3 Forgery Attacks
 - 13.3.4 Injection Attacks
 - 13.3.5 Directory Traversal and Command Injection Attacks
 - 13.3.6 URL Analysis
 - 13.3.7 Web Server Logs
- 13.4 Summary
- 13.5 Lesson Practice

14.0 Summarize Security Governance Concepts

- 14.1 Policies, Standards, and Procedures
 - 14.1.1 Policies
 - 14.1.2 Procedures
 - 14.1.3 Standards
 - 14.1.4 Legal Environment
 - 14.1.5 Governance and Accountability
- 14.2 Change Management
 - 14.2.1 Change Management Programs
 - 14.2.2 Allowed and Blocked Changes
 - 14.2.3 Restarts, Dependencies, and Downtime
 - 14.2.4 Documentation and Version Control
- 14.3 Automation and Orchestration
 - 14.3.1 Automation and Scripting
 - 14.3.2 Automation and Orchestration Implementation
- 14.4 Summary
- 14.5 Lesson Practice

15.0 Explain Risk Management Processes

- 15.1 Risk Management Processes and Concepts
 - 15.1.1 Risk Identification and Assessment
 - 15.1.2 Risk Management Strategies
 - 15.1.3 Risk Management Processes
 - 15.1.4 Business Impact Analysis
- 15.2 Vendor Management Concepts
 - 15.2.1 Vendor Selection
 - 15.2.2 Vendor Assessment Methods
 - 15.2.3 Legal Agreements
- 15.3 Audits and Assessments
 - 15.3.1 Attestation and Assessments
 - 15.3.2 Penetration Testing
 - 15.3.3 Exercise Types
- 15.4 Summary
- 15.5 Lesson Practice

16.0 Summarize Data Protection and Compliance Concepts

- 16.1 Data Classification and Compliance
 - 16.1.1 Data Types
 - 16.1.2 Data Classifications
 - 16.1.3 Data Sovereignty and Geographical Considerations
 - 16.1.4 Privacy Data
 - 16.1.5 Privacy Breaches and Data Breaches
 - 16.1.6 Compliance
 - 16.1.7 Monitoring and Reporting
 - 16.1.8 Data Protection
 - 16.1.9 Data Loss Prevention
- 16.2 Personnel Policies
 - 16.2.1 Conduct Policies
 - 16.2.2 User and Role-Based Training
 - 16.2.3 Training Topics and Techniques
 - 16.2.4 Security Awareness Training Lifecycle
- 16.3 Summary
- 16.4 Lesson Practice